Lectures on Topology and Physics

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Abstract

These are lectures about the relation between topology and physics given in the winter of 2016, in the Universidade Federal de Lavras. In the first lecture,....... The prerequisites are, essentially, a basic knowledge on the classical algebraic structures (groups and vector spaces) and some familiarity with the concepts from classical mechanics (like energy, linear momentum and Newton's laws of motion).

Contents

References

We list here some references which were used in the preparation of the lectures. The order in which they appear below corresponds more or less to the order in which the lectures were given. The first two are classical references about category theory and algebraic topology. The third and the fourth also are classical references. They cover Morse theory and their applications, including a prove of the generalized PoincarÃ©'s conjecture. The references of number 5-8 are about general aspects of geometry. The last three references present mathematical aspects of quantum field theories.

1. Mac Lane, S., *Categories for the Working Mathematician*, Springer, 1997.

2. May, J. P., *A Concise Course in Algebraic Topology*, Chicago University Press, 1999.

3. Milnor, J. W., *Morse Theory*, Princeton University Press, 1968.

4. Milnor, J. W., *Lectures on the h-Cobordism Theorem*, Princeton University Press, 1965.

5. Kobayashi S., Nomizu, K., *Foundations of Differential Geometry*, vol. 1, John Wiley & Sons, 1963.

6. Sternberg, S., *Lectures on Differential Geometry*, Prentice-Hall, 1964.

7. Dadam, F., Martins, Y. X., *Topologia, Geometria e Buracos Negros*, NEA, 2016.

8. Hori, K., et al, *Mirror Symmetry I*, AMS, 2003.

9. Deligne, P. et al, *Quantum Fields and Strings: A Course for Mathematicians*, vol. I, II, AMS, 1999.

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1 Language

Mathematics

There are two kinds of math: *naive* (or *intuitive*) *math* and *axiomatic* (or *rigorous*) *math*. In the naive math, the fundamental objects are assumed *primitives*. On the other hand, in the axiomatic math, these objects are defined by more elementary structures.

- **Example:** we have *naive set theory* and *axiomatic set theory*.

- independently of the case (naive or axiomatic) we need a *background language* (also called *logic*) to develop the theory.

- **Example:** in set theory, the language is *classical logic*.

- there is a natural language when we need develop a theory that makes connections between among different disciplines of math: the *categorical language*. Indeed, to give a *category* is just the same that specifying an area of math.

- More precisely, a category is an entity composed of three kinds of information: *objects*, *mappings* (also called *morphisms*) between objects and an associative *composition* of mappings. We also require the existence of *identity morphisms*.

- So, a category theory is just an entity in that we can talk about *commutative diagrams*.

- **Example:** there is a category **Set**, whose objects are sets, whose morphisms are exactly functions between sets and composition is the usual. The are of math specified by **Set** is just *set theory*.

- **Example:** the most of the categories that we will study are *concretes*. This means that objects are sets with additional structures, morphisms are functions between the underlying sets preserving additional structures and composition is usual.

- **Example:** we have *algebraic categories* (generically denoted by **Alg**). Objects are sets with additional operations and morphisms are functions that preserve this operations (is this case, they are called *homomorphisms*). So, we have the category **Grp** of groups, the category **Rng** of rings, the category $\mathbf{Mod}_R$ of R-modules, the category $\mathbf{Vec}_{\mathbb{K}}$ of vectors spaces, etc. The correspondents areas of math are group theory, ring theory, module theory and linear algebra.

- **Example:** We also have a category **Calc** that codify the differential calculus in one real variable. Indeed, their objects are the intervals $I \subset \mathbb{R}$ and the morphisms is just differentiable functions.

- Now, to see that the categorical language is really natural to talk about connections between areas of math, observe that the action "to specify an area of math" is itself an area of math!! This means that we can talk about mappings between different categories (and therefore, between different areas of math). This mappings are called *functors*. On another words, there is a category **Cat**, whose objects are categories and whose morphisms are functors. The area of math delimited by **Cat** is just *category theory*.

- But, *what is a functor*? To define it, remember that in general a mapping is a kind of rule that preserve all structures. A category is an entity with objects, morphisms, composition and identities. So, a functor between $\mathbf{C}$ and $\mathbf{D}$ just be a rule $F : \mathbf{C} \to \mathbf{D}$ that take objects and morphisms of $\mathbf{C}$ and return objects and morphisms of $\mathbf{D}$. Furthermore, F must satisfy $F(g \circ f) = F(g) \circ F(f)$ and $F(id_X) = id_{F(X)}$.

- Thus, all that we have discussed can be summarized in the following diagram:

$$\begin{tikzcd}
 & \textbf{naive math} \\
\begin{array}{c}\textbf{background language}\\ \text{(category theory)}\end{array} \arrow[ur, Rightarrow] \arrow[dr, Rightarrow] & \\
 & \textbf{axiomatic math}
\end{tikzcd}$$

- Now we observe that when doing math, a mathematician don't use rigorous arguments. Indeed, is only after make intuitions and reflections about some result that he uses of the all rigorous necessary to valid it. So, we can say that *naive math produces conjectures* and *rigorous math produces theorems*. Furthermore, naive math comes before of axiomatic math. With effect, to pass of naive math to rigorous math is necessary fix a model in which the conjecture is so proven.

$$\begin{tikzcd}
 & \textbf{naive math} \arrow[r, Rightarrow, "\text{produces}"] & \textbf{conjectures} \arrow[dd, Rightarrow, "\text{model}"] \\
\begin{array}{c}\textbf{background language}\\ \text{(category theory)}\end{array} \arrow[ur, Rightarrow] \arrow[dr, Rightarrow] & & \\
 & \textbf{axiomatic math} \arrow[r, Rightarrow, "\text{produces}"'] & \textbf{theorems}
\end{tikzcd}$$

- These models in general are given by algebraic or geometric framework. Thus, we can say that we *think using category theory and we do using geometry and algebra.*

Physics

- In physics, the situation is totally different. Indeed, in physics we have a restriction about the *existence*. More precisely, we have a connection with *ontology*. The connection is given by the *empiricism*. So, more precisely, *although math is a strictly logical discipline, physics is logical and ontological.*

- This restriction produces many difficulties. For example, to consider a sentence as true, the logical consistence is not sufficient: we also need ontological consistence. So, *even tough a sentence be logical consistent, to be considered true it must be consistent with all possible experiment!* This also can be expressed in terms of a commutative diagram:

$$\begin{tikzcd}
\textbf{phylosophy} \arrow[r, Rightarrow] \arrow[d, Rightarrow] & \textbf{logic} \arrow[d, Rightarrow] \\
\textbf{ontology} \arrow[r, Rightarrow, "\text{empiricism}"'] & \textbf{physics}
\end{tikzcd}$$

- But, remember that logic is the language to construct the commutative diagram of math. So, joint the last diagram with the math diagram, we have a new commutative diagram:

$$\begin{tikzcd}
 & \textbf{conjectures} \arrow[r, Rightarrow] & \textbf{theorems} \\
 & \begin{array}{c}\textbf{naive}\\ \textbf{math}\end{array} \arrow[u, Rightarrow] & \\
\textbf{phylosophy} \arrow[r, Rightarrow] \arrow[d, Rightarrow] & \textbf{logic} \arrow[r, Rightarrow] \arrow[u, Rightarrow] \arrow[d, Rightarrow] & \begin{array}{c}\textbf{axiomatic}\\ \textbf{math}\end{array} \arrow[uu, Rightarrow] \\
\textbf{ontology} \arrow[r, Rightarrow, "\text{empiricism}"'] & \textbf{physics} &
\end{tikzcd}$$

- the most important fact that concerns the relation between physics and mathematics is the following: in the previous diagram, the arrow

$$\textbf{logic} \Longrightarrow \textbf{physics} \qquad \text{has an inverse} \qquad \textbf{physics} \Longrightarrow \textbf{logic}$$

- It is called *physical insight*. So, the composition of *physical insight* with axiomatic math produces a new arrow called *mathematical physics*.

$$\begin{tikzcd}
\textbf{logic} \arrow[r, Rightarrow] & \textbf{axiomatic math} \arrow[dl, Rightarrow, "\text{math. phys.}"] \\
\textbf{physics} \arrow[u, Rightarrow, "\text{insight}"] &
\end{tikzcd}$$

- The full relation between physics and mathematics is thus given by

$$\begin{tikzcd}
 & \textbf{conjectures} \arrow[r, Rightarrow] & \textbf{theorems} \\
 & \textbf{naive math} \arrow[u, Rightarrow] & \\
\textbf{phylosophy} \arrow[r, Rightarrow] \arrow[d, Rightarrow] & \textbf{logic} \arrow[r, Rightarrow] \arrow[u, Rightarrow] \arrow[d, Leftrightarrow] & \textbf{axiomatic math} \arrow[uu, Rightarrow] \arrow[dl, Rightarrow] \\
\textbf{ontology} \arrow[r, Rightarrow, "\text{empiricism}"'] & \textbf{physics} &
\end{tikzcd}$$

- **Conclusion:** *we can use physical insight to do naive math and, therefore, to produce conjectures. This conjectures can eventually proven, getting theorems!!! Furthermore, with theorems on hands, we can create mathematical models to physics (mathematical physics) which must be verified experimentally.*

- Now we can say that the major objective is the following: study some examples of the following cycle:

$$\begin{tikzcd}
\textbf{physics} \arrow[r, Rightarrow] & \textbf{logic} \arrow[r, Rightarrow] & \textbf{naive math} \arrow[r, Rightarrow] & \textbf{conjectures} \arrow[r, Rightarrow] & \textbf{theorems} \arrow[llll, Rightarrow, bend left]
\end{tikzcd}$$

Abstraction

- Consider the following problems:
  - unify math;
  - unify physics.

- how math is determined only by logic, *to unify math we* ***need*** *a very general background language*. On the other hand, physics is partially determined by logic. Its suggest that *more abstract and general background language* ***help us*** *to unify physics.*

- We have discussed that category theory is a natural language. In this subsection, we view explicitly that it is *really abstract*! More precisely, we will see that many concepts of different areas of math admits an abstraction using the categorical language. Indeed, we will that they are particular cases of the categorical concept of *limit of a functor*.

- Let $F : \mathbf{J} \to \mathbf{C}$ be a functor and $A \in \mathbf{C}$ and object. We define a *cone of F with vertex on A* as being the collection of all diagrams like this:

$$\begin{tikzcd} & F(X) \arrow[dd, "F(f)"] \\ A \arrow[ur, bend left] \arrow[dr, bend right] & \\ & F(Y) \end{tikzcd}$$

- Consider two cones of $\mathbf{C}$ (say with vertex on A and B). We say that the cone on B *collapse on* the cone on A when there exists an *unique* map $u : B \to A$ that makes commutative every diagram like the below.

$$\begin{tikzcd} & & F(X) \arrow[dd, "F(f)"] \\ B \arrow[r, dashed, "u"] \arrow[urr, bend left] \arrow[drr, bend right] & A \arrow[ur, bend left] \arrow[dr, bend right] & \\ & & F(Y) \end{tikzcd}$$

- The *limit* of a functor F is a cone which is *universal with respect to the property of collapsing.* In other words, is a cone in which each other cone collapse.

- Not every functor have a limit. We will see examples along the others seminars. On the other hand, if exists, the limit of a functor is *unique module isomorphisms of your vertex.* This happens because we define the collapse of a cone on another cone as being an *unique* map $u : B \to A$.

- In the following, we will see some examples of limits. The different classes will be characterized by the domain $\mathbf{J}$ of the functor $F : \mathbf{J} \to \mathbf{C}$.

- In general, we consider $\mathbf{J}$ as being a *finite category.* This means that it have a finite quantity of objects and of morphisms linking them. So, all information contained on $\mathbf{J}$ can be compactified on an unique diagram and a functor $F : \mathbf{J} \to \mathbf{C}$ is just a indexed copy of the diagram that characterizes **J** inside **C**, like this:

$$\begin{tikzcd}
1 \arrow[r] \arrow[d] & 2 \arrow[d] \arrow[dr] & \\
3 \arrow[u, bend left] \arrow[ur] & 4 \arrow[l] \arrow[r] & 5
\end{tikzcd}
\qquad
\begin{tikzcd}
X_1 \arrow[r] \arrow[d] & X_2 \arrow[d] \arrow[dr] & \\
X_3 \arrow[u, bend left] \arrow[ur] & X_4 \arrow[l] \arrow[r] & X_5
\end{tikzcd}$$

Products

- The first class of limits that will study are the *binary products*. They are the limit of functors defined on the category **J** that have only two objects $1, 2$ and whose morphisms are only identities. Then a functor $F : \mathbf{J} \to \mathbf{C}$ is a rule that label two objects on **C** (say X_1 and X_2). How there are not morphisms $1 \to 2$, a cone of F with vertex on $A \in \mathbf{C}$ is a diagram like this:

$$\begin{tikzcd}
 & X_1 \\
A \arrow[ur, bend left, "pr_1"] \arrow[dr, bend right, "pr_2"'] & \\
 & X_2
\end{tikzcd}$$

- So, we can say that a cone of F is just a way to project a certain object A into X_1 and into X_2. Therefore, the limit of F is an object that can be projected into the indexed objects X_1 and X_2 of an *universal way*. This means that, if $pr'_i : B \to X_i$ is another maps, then there exists an unique $u : B \to A$ that makes commutative the diagram below.

$$\begin{tikzcd}
 & & X_1 \\
B \arrow[r, dashed, "u"] \arrow[urr, bend left, "pr'_1"] \arrow[drr, bend right, "pr'_2"'] & A \arrow[ur, bend left, "pr_1"] \arrow[dr, bend right, "pr_2"'] & \\
 & & X_2
\end{tikzcd}$$

- A category in which every indexed pair of objects admits universal space (that is, in which every functor $F : \mathbf{J} \to \mathbf{C}$) is saw to have *binary products*.

- **Example:** the category **Set** have binary products. Indeed, given sets X_1 and X_2, we can form the cartesian product $X_1 \times X_2$ between then and consider pr_1 and pr_2 as the projections on first and second part.

- **Example:** The category $\mathbf{Vec}_{\mathbb{K}}$ of vector spaces also have binary products. They are the cartesian product of the underlying sets endowed with the usual product structure. The same is also valid for every algebraic category, like **Grp** or **Rng**.

- **Example:** the category **Calc** has not binary products. The proof is not difficult, but is not a immediate fact. The idea is the following: an interval is an one dimensional object. On the other hand, their product must be a two dimensional object and, therefore, cannot be an interval.

- We observe that the same analysis can be applied for every category $\mathbf{J}$ which have an arbitrary quantity of objects, but whose morphisms are only identities. Indeed, in this case a functor $F : \mathbf{J} \to \mathbf{C}$ is a family of objects $X_i \in \mathbf{C}$ indexed by the set of objects of $\mathbf{J}$. Analogously, a cone of F is formed by a family of maps $\pi_i : A \to X_i$ and the limit of F is just an universal cone. That is, one such that for every family of maps $\pi_i' : B \to X_i$ there is an unique collapsing map $u : B \to A$.

- A *category with arbitrary products* is one in which every functor $F : \mathbf{J} \to \mathbf{C}$, define on a category like above, have limits.

- **Example:** Suppose that $\mathbf{J}$ have a *vacuum set of objects*. This means that any functor $F : \mathbf{J} \to \mathbf{C}$ does nothing. Therefore, a cone is composed only by the vertex. So, the limit is an object A such that for every another object B there exists an unique morphism $B \to A$.

Equalisers

- Now, we study another class of limits: the *equalisers*. They correspond to the limits of functors defined on the category with have only two objects and only two parallel morphisms between them. Indeed, this category can be codified in the below diagram, of mode that a functor $F : \mathbf{J} \to \mathbf{C}$ is just a pair of parallel arrows of $\mathbf{C}$:

$$1 \rightrightarrows 2 \qquad\qquad X_1 \overset{f}{\underset{g}{\rightrightarrows}} X_2$$

- Furthermore, a cone of F is given by the following commutative diagrams, which can be joined on an unique diagram:

$$\begin{tikzcd} & X_1 \arrow[dd, "f"] \\ A \arrow[ur, bend left, "\imath"] \arrow[dr, bend right, "\jmath"'] & \\ & X_2 \end{tikzcd} \qquad \begin{tikzcd} & X_1 \arrow[dd, "g"] \\ A \arrow[ur, bend left, "\imath"] \arrow[dr, bend right, "\jmath"'] & \\ & X_2 \end{tikzcd} \qquad \begin{tikzcd} & X_1 \arrow[dd, shift right, "f"'] \arrow[dd, shift left, "g"] \\ A \arrow[ur, bend left, "\imath"] \arrow[dr, bend right, "\jmath"'] & \\ & X_2 \end{tikzcd}$$

- In more precise words, a cone of F is just a map $\imath : A \to X_1$ such that $f \circ \imath = g \circ \imath$. So, the limit of F (called the *equaliser between f and g*) is an universal cone $\imath : A \to X_1$ in the sense that, if $\imath' : B \to X_1$ is another map satisfying $f \circ \imath' = g \circ \imath'$, then exists an unique $u : B \to A$ such that $\imath \circ u = \imath'$.

- A *category with equalisers* is one in which every pair of morphisms have equaliser.

- **Example:** the category **Set** have equalisers. Indeed, given $f, g : X_1 \to X_2$, to construct the equaliser between then, we need first get a cone. That it, a map $\imath : A \to X_1$ such that $f \circ \imath = g \circ \imath$. Evidently, if A is the subset of every $x \in X_1$ such that $f(x) = g(x)$, then the inclusion satisfy the require commutative property. We affirm that it really is the equaliser between f and g. We need proof that it is universal. So, let $\imath' : B \to X_1$ satisfying $f \circ \imath' = g \circ \imath'$. Thus $f(\imath'(x)) = g(\imath'(x))$. This means that f and g is equal when restricted to the set $\imath'(X)$. But A is the minor set in with $f = g$. So, $\imath(X_1) \subset \imath'(X_1)$, of mode that $\imath : A \to X_1$ is really universal.

- **Example:** every algebraic category has equalisers. Following the idea of the previous example, given $f, g : X_1 \to X_2$ we need search for the minor object in which $f = g$. But $f = g$ iff $f - g = 0$. So, this object is just $\ker(f - g)$.

Completeness

- Now, we look for a third class of limits: the *pullbacks*. They are the limits of functors defined on the category that can be codified on the below diagram.

$$
\begin{tikzcd}
 & 2 \arrow[d] \\
1 \arrow[r] & 0
\end{tikzcd}
\qquad\qquad
\begin{tikzcd}
 & X_2 \arrow[d, "g"'] \\
X_1 \arrow[r, "f"'] & Y
\end{tikzcd}
$$

- The functors $F : \mathbf{J} \to \mathbf{C}$ are in bijective correspondence with the diagrams like ones the present above. Analogously to the previous examples, we can see that a cone of F is just an object A with arrows $\pi_i : A \to X_i$ such that $f \circ \pi_1 = g \circ \pi_2$. In other words, that makes commutative the following square.

$$
\begin{tikzcd}
A \arrow[r, "\pi_2"] \arrow[d, "\pi_1"'] & X_2 \arrow[d, "g"] \\
X_1 \arrow[r, "f"'] & Y
\end{tikzcd}
$$

- The limit of F (called the *pullback between f and g*) is then an universal commutative square. This means that, if $\pi_i' : B \to X_i$ is another pair of maps such that $f \circ \pi_1' = g \circ \pi_2'$, then there exists an unique $u : B \to A$ making commutative the below diagram.

$$\begin{tikzcd}
B \arrow[dr, dashed, "u"] \arrow[drr, bend left, "\pi_1'"] \arrow[ddr, bend right, "\pi_2'"'] & & \\
& A \arrow[r, "\pi_2"] \arrow[d, "\pi_1"'] & X_2 \arrow[d, "g"] \\
& X_1 \arrow[r, "f"'] & Y
\end{tikzcd}$$

- A *category with pullbacks* is just one in which every pair of morphisms have pullback. Observe that the pullback of f and g is a certain kind of "binary product that depend of f and g". But the equaliser $eq(f, g)$ is a limit that depend of f and g. This motivate the following

- **Affirmative:** *if a category has binary products and equalisers, so it also has pullbacks.*

  - let $f : X_1 \to Y$ and $g : X_2 \to Y$ two morphisms. Consider the following square:

$$\begin{tikzcd}
X_1 \times X_2 \arrow[r, "pr_2"] \arrow[d, "pr_1"'] & X_2 \arrow[d, "g"] \\
X_1 \arrow[r, "f"'] & Y
\end{tikzcd}$$

    It is not commutative and is not universal. But, to turn it commutative and universal, enough replace $X_1 \times X_2$ by the equaliser between $f \circ pr_1$ and $g \circ pr_2$

- **Conclusion:** *Different limits can be dependently. Indeed, with binary products and equalisers, we can construct pullbacks. But, much more is valid: can be prove that* ***every*** *limit can be constructed by arbitrary products and equalisers.*

- A category $\mathbf{C}$ in which every functor $F : \mathbf{J} \to \mathbf{C}$ have limit is called *complete*. So, the previous conclusion show that *a category is complete iff has products and equalisers.*

- **Example:** the category **Set**, as well every algebraic category, is complete.

Duality

- Remember that a category is an entity in which we can talk about commutative diagrams. But diagrams are composed by commutative arrows. Now, given a commutative diagram, we can get another simply inverting the direction of the arrows. But the arrows on a diagram corresponds to the morphisms of the underlying category. So, given a category $\mathbf{C}$, we can construct another category $\mathbf{C}^{op}$ (called the *opposite of* $\mathbf{C}$) simply inverting the direction of the morphisms.

- In more precise words, $\mathbf{C}^{op}$ is the category such that
  - have the same objects of $\mathbf{C}$;
  - for every $f : X \to Y$ on $\mathbf{C}$ there exists an unique $f^{op} : Y \to X$ on $\mathbf{C}$;
  - the composition $g^{op} \circ f^{op}$ is defined by $(f \circ g)^{op}$.
- Remember that functors preserves commutative diagrams. So, for every $F : \mathbf{C} \to \mathbf{D}$ there exists another functor $F^{op} : \mathbf{C}^{op} \to \mathbf{D}^{op}$, such that
$$F^{op}(X) = F(X) \quad \text{and} \quad F^{op}(f^{op}) = F(f)^{op}.$$
- This means that, in fact, the operation "to take the opposite" defines a functor
$$(-)^{op} : \mathbf{Cat} \to \mathbf{Cat},$$
characterizing a duality principle: *all that can be defined and proved using only commutative diagrams on* $\mathbf{C}$ *also can be defined and proved on* $\mathbf{C}^{op}$.

Classification

- the classification problem.

Problems

1. A *groupoid* is just a category whose morphism are always isomorphisms. Show that we have a category $\mathbf{Grp}$ whose objects are groupoids and whose morphisms are functors. Show that exists a functor $(-)_{pd} : \mathbf{Cat} \to \mathbf{Grpd}$ that assign to every category $\mathbf{C}$ the correspondent category $\mathbf{C}_{pd}$ obtained from $\mathbf{C}$ subtracting every morphism that is not an isomorphism.
2. Show that there exists two natural functor $\mathbf{Grp} \to \mathbf{Cat}$. The first assign to every group G a category whose an unique object and the second with only morphisms are the identities. Show that, in fact, these functors take values on the category of groupoids.
3. Given a functor F, show that we have a category $\mathrm{Cone}(F)$ whose objects are cones and whose morphisms are collapsing maps. Show that a limit of F is just a initial object of $\mathrm{Cone}(F)$. Apply the duality principle to conclude that also exists a category $\mathrm{CCone}(F)$ whose objects are cocones of F and such that the colmits of F are in bijection with the final objects of $\mathrm{CCone}(F)$.
4. Prove the uniqueness of limits and colimits.
5. adjunctions. Prove that $\imath : \mathbf{Mod}_R \to \mathbf{Set}$ have adjoint. This means that we can talk about modules freely generated by sets.

2 Topology

- Topology is the area of math in which we can discern *local properties* of *global properties*. Indeed, we say that a property is *local* when it is valid in the *neighborhood* of any point. On the other hand, a property is *global* when it is valid in the whole space. So, *neighborhood* is the fundamental concept of topology

- Evidently, global properties is more strong than local properties. With this in mint, given a local property, we would like extend it to a global property. A way to do this is using the notion of *compatible neighborhood*. Indeed, suppose that a local property is valid on the neighborhood of two different points. So, if the neighborhoods are compatible, then the property will be valid on the union of them.

- Thus, in topology, we usually need to talk about *compatible neighborhoods*. In the following, we will formalize this idea.

- Let X a set and $x \in X$ and element. A *neighborhood* of x is just a subset of X that contains x. A *topology* on X is a map τ, usually called *basis for the topology*, that assign a set of *fundamental neighborhoods* $\tau(x)$ for every $x \in X$, which are compatible in the following sense:

  - given $B(x) \in \tau(x)$ and $B(y) \in \tau(y)$, then for every $z \in B(x) \cap B(y)$ their exists $B(z) \in \tau(z)$ entirely contained on the intersection $B(x) \cap B(y)$. This means, in particular, that every x have arbitrarily small fundamental neighborhoods.

- A set endowed with a basis is called a *topological space*. So, we can say that *topology is the study of the topological spaces*.

- If τ is a basis on a topological space X, then a τ-*open* subset of X is one which can be written as the union of fundamental neighborhoods (possibly of different points). Different basis can produce the same collection of open subsets. This means that *the concept of open subsets is more important than the concept of neighborhood*.

- **Example:** We have a basis of $\mathbb{R}^n$ given by the rule that assign to every $x \in \mathbb{R}^n$ the set $\tau(x)$ of open balls $B_r(x)$ centered on x.

- **Example:** Every subset $A \subset X$ of a topological space X admits a natural induced topological structure. Indeed, if τ is a basis of X, so we can take the rule $\tau|_A$ that assign to every $x \in X$ the set $\tau|_A(x)$ of all intersections $B(x) \cap A$.

- **Example:** Important particular cases of the previous examples are the sphere $\mathbb{S}^{n-1}$, the disk $\mathbb{D}^n$ and the semispace $\mathbb{H}^n$, all considered as subspaces of $\mathbb{R}^n$.

Category

- Remember that there is a correspondence between areas of math and categories. Topology is an area of math that study topological spaces. So, must be exists a category **Top** whose objects are topological spaces. Let construct it.

- A topological space is just a set with the notion of neighborhood. Thus, is a set endowed with an additional structure. Thus, the idea is consider **Top** as a concrete category. This means that, if X and Y are topological spaces, the morphisms will be functions between the underlying sets that preserves neighborhoods.

- More precisely, a morphism $f : X \to Y$ (called *continuous map*) is a function with the following property:

  - for every fundamental neighborhood $B(f(x))$ there exists $B(x)$ such that, if $y \in B(x)$, then $f(y) \in B(f(x))$.

- We can rapidity check that the composition of continuous functions also is continuous, of mode that, having then as morphisms, **Top** is a well defined concrete category.

- **Example:** the previous notion of continuous function agree with the notion of continuity present on the curses of analysis. Indeed, consider on $\mathbb{R}^n$ the basis given by open balls. So, the continuity of $f : \mathbb{R}^n \to \mathbb{R}^m$ can be translated in the following way:

  - for every ball $B_\varepsilon(f(x))$ there exists $B_\delta(x)$ such that, if $y \in B_\delta(x)$, then $f(y) \in B_\varepsilon(f(x))$. That is, such that if

$$\|y - x\| < \delta, \quad \text{then} \quad \|f(y) - f(x)\| < \varepsilon.$$

Completeness

- We affirm that **Top** is complete and cocomplete. By the existence theorem of limits, we only need check that **Top** has products and equalisers. Furthermore, by duality, to prove cocompleteness, we only construct coproducts and coequalizers.

- **products:** we will construct binary products. The general case will be given analogously. Let X_1 and X_2 two topological spaces. Given basis τ_1 and τ_2, we consider on the cartesian product $X_1 \times X_2$ the rule $\tau_1 \times \tau_2$ that assign to every (x_1, x_2) the set of all $B(x_1) \times B(x_2)$ with $B(x_i) \in \tau(x_i)$. It is a basis in which the projections pr_i are continuous function. Thus, to prove that $\tau_1 \times \tau_2$ really turn $X_1 \times X_2$ in the binary product on **Top**, enough verify that it is universal. This really is the case, because every another basis that makes pr_i continuous must contain $\tau_1 \times \tau_2$.

- **equalisers:** given continuous functions $f, g : X \to Y$, let $eq(f, g)$ be the equaliser between f and g on **Set**. So, $eq(f, g) \subset X$. With the subspace topology, the inclusion is continuous, and then it is a natural candidate of equaliser on **Top**. It really is, because the subspace topology is the minor topology that make inclusions be continuous.

- **coproducts:** let X_1 and X_2 be topological spaces and consider their coproduct on **Set**. There is, consider the disjoint union $X_1 \sqcup X_2$. If τ_1 and τ_2 are basis, take the rule $\tau_1 \sqcup \tau_2$ such that $\tau_1 \sqcup \tau_2(x) = \tau_i(x)$ when $x \in X_i$. Then the inclusions $\imath_i$ are continuous, of mode that $X_1 \sqcup X_2$, endowed with $\tau_1 \sqcup \tau_2$, is a candidate of coproduct on **Top**. It really is the coproduct. Indeed, each other topology that makes $\imath_i$ continuous contain $\tau_1 \sqcup \tau_2$.

- to construct equalisers, we will use the following

  - **Fact:** let X be a topological space, Y an arbitrary set and $\pi : X \to Y$ a function. If τ is a basis on X, then the rule τ_π, such that $\tau_\pi(y)$ is the union of every $\tau(x)$, with $\pi(x) = y$, is the minor basis on Y that makes π continuous.

- **coequaliser:** let $f, g : X \to Y$ be continuous function and let $\text{coeq}(f, g)$ be their coequaliser on **Set**. It is the quotient space of X by the relation that collapse on an unique point every x in which $f(x) = g(x)$. So, we have a map $\pi : X \to \text{coeq}(f, g)$. Let τ a basis on X and consider the basis τ_π on $\text{coeq}(f, g)$. By the previous fact, it is universal with respect to the property of turn π continuous. So, with τ_π, the coequaliser on **Set** is a coequaliser on **Top**.

Classification

- Here we will consider the *classification problem of* **Top**. This means that we will try decide which is the objects of **Top** module isomorphisms. So, first of all, we need become more familiar with the isomorphisms of such category.

- Remember that in an arbitrary category **C**, two objects X, Y are called *isomorphics* when exists morphisms $f : X \to Y$ and $g : Y \to X$ such that the compositions $g \circ f$ and $f \circ g$ are identities.

- So, in **Top**, two topological spaces will be isomorphics (usually called *homeomorphics*) when we have continuous functions f and g between then, called *homeomorphisms*, such that $g \circ f$ and $f \circ g$ are identities. Intuitively, this means that X can be continuously deformed on Y without lacerate, punch and collapse dimensions.

- **Example:** we expect that every open ball on $\mathbb{R}^n$ is homeomorphic to each other. Indeed, we can transform one on another simply translating and rescaling. Indeed, given $B_r(p)$ and $B_s(q)$, we first consider the translation $x \mapsto x + q - p$. So, we compose it with the scaling $x \mapsto \frac{s}{r} x$.

- **Example:** on the other hand, we don't expect that some $B_r(p)$ be homeomorphic to the point p. Indeed, to pass continuously of a ball to a point we need collapse dimension. With the same argument, we don't expect that balls of different euclidean spaces be homeomorphics.
- **Example:** we don't expect that the cylinder $\mathbb{S}^1 \times \mathbb{R}$ be homeomorphic to $\mathbb{S}^1$, because to construct a deformation of one to another, we need collapse dimension. We also don't expect the existence of a homemorphism between $\mathbb{R}^n - 0$ and $\mathbb{R}^n$, because to construct $\mathbb{R}^n - 0$ we need withdraw a point, which is the same that punching. Finally, we don't expect that the torus $\mathbb{S}^1 \times \mathbb{S}^1$ be homeomorphic to the sphere $\mathbb{S}^2$, because, differently of the sphere, the torus have a handle.

Invariants

- Remember that the idea of look for homeomorphic as continuous deformations is *merely intuitive*. This justify the usage of "to expect" on the previous examples.
- Indeed, if we need proof that two topological spaces are really homeomorphics, we need construct a homeomorphim between them. For example, on the proof that all balls are homeomorphics, first was inferred the existence of a homeomorphism, but then it was actually build. But constructive proofs in general are very hard to do. So, given two topological spaces, the idea is try to prove that one cannot be homeomorphic to each other. As we have discussed on the first lecture, the strategy is focus on the construction of invariants.
- Remember that an *invariant* of a category $\mathbf{C}$ is just a property $\mathcal{P}$ such that, if $\mathcal{P}$ is valid on a certain X, then it must be valid on each Y isomorphic to X. So, if there exists an invariant which is valid on X but not on Y, then these objects cannot be isomorphics.
- There are many invariants of **Top** that can be build simply imposing restrictions on the quantity of open subsets of the space. The most commons are *compactness* and the *Haussdorf property*. They are respectively related to the existence of few or many opens.
- Occurs that this invariants is not much strong and, therefore, they not help on the classification problem of **Top**. For example, they are not sufficient to prove that $\mathbb{S}^1 \times \mathbb{S}^1$ and $\mathbb{S}^2$ are not homeomorphics. So, the idea is search for *more strong topological invariants*.
- How proceed? Remember that for every functor $F : \mathbf{C} \to \mathbf{D}$ corresponds an invariant of $\mathbf{C}$. Therefore, the strategy is search for functors from **Top** to another category. The *algebraic topology* is the area of topology that work on the construction of functors $F : \mathbf{Top} \to \mathbf{Alg}$. This is interesting because, is general, classify algebraic structures is more simple than classify topological spaces.

Homology

- There is a simple strategy can be use to construct not only one, but a *sequence of topological invariants!* The idea is the following:

- let $\mathbf{C}$ be an algebraic category. Thus, on $\mathbf{C}$ we can talk about kernels, images and quotients by kernels (cokernels). In particular, given $\partial_i : X_i \to X_{i-1}$ and $\partial_{i+1} : X_{i+1} \to X_i$, suppose that

$$\ker(\partial_{i+1}) \subset \operatorname{img}(\partial_i). \tag{1}$$

  Then we can take the quotient

$$\operatorname{img}(\partial_{i+1}) / \ker(\partial_i), \tag{2}$$

  which is a member of $\mathbf{C}$.

- Now, suppose that we have not only two maps, but a sequence X_* of morphisms ∂_i of $\mathbf{C}$ satisfying (1). Thus we can construct a sequence of quotients like (2).

- We can construct a category $\mathbf{Ch_C}$ of such sequences. They are called *chain complexes*. A morphism of chain complex is just a sequence h_* of maps $h_i : X_i \to X'_i$ that commutes with ∂_i and ∂'_i. This maps pass to quotient and induces morphisms

$$(h_i)_* : \operatorname{img}(\partial_{i+1}) / \ker(\partial_i) \to \operatorname{img}(\partial'_{i+1}) / \ker(\partial'_i).$$

- This means that exists a sequence of functors $H_i : \mathbf{Ch_C} \to \mathbf{C}$ such that

$$H_i(X_*) = \operatorname{img}(\partial_{i+1}) / \ker(\partial_i) \quad \text{and} \quad H_i(h_*) = (h_i)_*.$$

- Now, let $F : \mathbf{Top} \to \mathbf{Ch_C}$ be a functor such that to each topological space X assign a chain complex X_* and to each continuous function $f : X \to Y$ assign a chain map h_*. Thus, the compositions $F_i = H_i \circ F$ defines a family of invariants on $\mathbf{Top}$. We say that they are *homological invariants*.

- **Example:** homological invariants really exists. More precisely, let build a functor $F : \mathbf{Top} \to \mathbf{Ch_C}$.

$$C_n(X) = \{\sum a_i \cdot \sigma_i : \Delta_n \to X,\ a_i \in \mathbb{K}\}.$$

Cohomology

Calculability

- The following example show that, in general, construct invariants in not enough. Indeed, we also need tools to calculate them.

- **Example:** We have a functor Aut : $\mathbf{Top}_{pd} \to \mathbf{Grp}$ that assign to every topological space the group of your homeomorphisms. But in general we don't know identify this group.
- The most important tools are:
  - *preservation of limits and colimits.*
  - *existence of exact sequences.*

CW-Complexes

- the problem of calculability and the problem of unicity.
- CW-complex is a well behaved class of topological spaces with respect to these problems.
- uniqueness of (well behaved homology and cohomology theories) for CW-complexes.

Problems

- covering spaces. $\mathbb{R}$ is the cover of $\mathbb{S}^1$. $\mathbb{S}^n$ is the cover of $\mathbb{P}^n$.
- fundamental group.
- fundamental group of covering spaces. fundamental group of the circle.
- homotopy category.
- homotopy invariants. Show that π_1 is a homotopy invariant.
- homotopy limits and colimits. Show that products and coproduts are homotopy limits.
- show that π_1 preserve products. It also preserve coproduts?
- show that the projective space $\mathbb{P}^n$ admits a natural decomposition on CW-complex.

3 Manifolds

Modeling

- Here we look for the classification problem of **Top** in a different perspective. Indeed, *if classify general topological spaces is hard, we will restrict our attention on the topological spaces which are assumed locally classified. Then, we will try globalize this classification.*

- The idea is the following: we fix a space X, called *model*, and look for the spaces locally equivalents to X. That is, in which every point admits a neighborhood homeomorphic to a neighborhood of X. So, for every p there exists a map $\varphi : U \to X$, defined on a neighborhood U of p, such that

  - $\varphi(U)$ is open on X;
  - the functions $\varphi : U \to \varphi(U)$ is a homeomorphism.

- Such maps are called *charts on* p. The image $\varphi(p)$ is the *representation of* p *on the chart* φ. The collection of charts is an *atlas.*

- **Example:** X is obviously modelled over yourself. Enough take $\varphi = id_X$.

- We have a category $\mathbf{Top}_X$, whose objects are spaces modelled over X and whose morphisms are just continuous maps. Two observations:

  - in general, $\mathbf{Top}_X \neq \mathbf{Top}$. This means that not every topological space can be modeled over a fixed space X;
  - more rigid spaces X generally produces a more restrict category $\mathbf{Top}_X$, but more easily to classify. On the other hand, few rigid spaces produces big categories with few properties and, then, hard to classify

- So, the idea is search for a nice model. For example, we can consider initially a very restrict model and then go weakening it slowly to get a category with more properties, as represented on the below diagram

$$
\begin{tikzcd}
\textbf{modeling over a certain space} \arrow[r, Rightarrow] & \textbf{category with few properties} \arrow[d, Rightarrow] \\
& \textbf{model space is rigid} \arrow[ul, Rightarrow, "\text{weake-ning}"]
\end{tikzcd}
$$

Manifolds

- Here we will ilustra the previous diagram.

- We start with the most simple nontrivial topological space: $\mathbb{R}^n$. The spaces modeled over them are the *n-dimensional manifolds*. So, if M is a n-manifold, then every p admits a neighborhood and a chard $\varphi : U \to \mathbb{R}^n$, in which we can write p as a list of real numbers:
$$\varphi(p) = (x_1, ..., x_n).$$
  - **Example:** the euclidean space $\mathbb{R}^n$ with a unique chart: $\varphi = id$.
  - **Example:** the sphere $\mathbb{S}^n$ with two charts: the stereographic projections.

- Now, we observe that $\mathbf{Top}_{\mathbb{R}^n}$ not contain some natural examples of spaces:
  - **Example:** the disk $\mathbb{D}^n$ is not a manifold:
  - **Example:** similarly, the semispace $\mathbb{H}^n$ is not a manifold.

- Furthermore, we also observe that $\mathbf{Top}_{\mathbb{R}^n}$ is not well behaved with respect to limits and colimits.
  - **Example:** we have products, but don't have pullbacks. Indeed, ...
  - **Example:** we have coproduts, but don't have pushouts. Indeed, gluing two lines along a point is not a manifold.

- The first problem occurs exactly because *a space M modelled over X must have the same types of open sets that X*. But $\mathbb{R}^n$ only one type, while $\mathbb{D}^n$ and $\mathbb{H}^n$ have two types. So, to include $\mathbb{D}^n$ and $\mathbb{H}^n$, we can work, for example, on $\mathbf{Top}_{\mathbb{H}^n}$. The objects of this category are the *manifolds with boundary*. We justify the name. Indeed, we have two functors:
$$\mathrm{int} : \mathbf{Top}_{\mathbb{H}^n} \to \mathbf{Top}_{\mathbb{R}^n} \quad \text{and} \quad \partial : \mathbf{Top}_{\mathbb{H}^n} \to \mathbf{Top}_{\mathbb{R}^{n-1}}.$$

- This resolves the first problem, but not the second. Indeed, $\mathbf{Top}_{\mathbb{H}^n}$ has not pushouts: gluing closed intervals along a point we have a space that not is manifold with boundary.

- Orbifolds and Diffeological spaces.

Conjugation

- Let $\mathbf{C}$ be an arbitrary category. We can construct a new category $\mathrm{Arr}(\mathbf{C})$ whose objects are the morphisms (the arrows) of $\mathbf{C}$. Indeed, the morphisms between $f, g \in \mathbf{C}$ are the *commutative squares*. That is, are maps $h, h' : f \to g$ such that $h' \circ f = g \circ h$.

- The isomorphisms of $\mathrm{Arr}(\mathbf{C})$ are called *conjugations*. They are exactly commutative squares in which h and h' are isomorphisms of $\mathbf{C}$.

- **Example:** on the category $\mathrm{Arr}(\mathbf{Vec}_{\mathbb{K}})$, the conjugations are just the conjugation of linear transformations.

- Now, we look for the category $\mathbf{Top}$. Remember that topology is the are of of in which we can differ *local things* of *global things*. So, we can build a local version $\mathrm{Loc}(\mathbf{Top})$ of $\mathrm{Arr}(\mathbf{Top})$, of mode that we can talk about *local conjugations*.

- Indeed, the objects are maps of $\mathbf{Top}$, but instead consider commutative squares as morphisms of $\mathrm{Loc}(\mathbf{Top})$, we take *local commutative square*. More precisely, a morphism between f and g is family $h_p, h'_{f(p)}$ of maps, respectively defined on the neighborhoods of each p and $f(p)$, such that $h'_{f(p)} \circ f = g \circ h_p$. The *local conjugations* are then the isomorphisms of $\mathrm{Loc}(\mathbf{Top})$. In other words, are such that the maps h_p and $h_{f(p)}$ are homeomorphisms over your images.

- For every model space X we can build, in a completely analogous way, the categories $\mathrm{Arr}(\mathbf{Top}_X)$ and $\mathrm{Loc}(\mathbf{Top}_X)$. So, we can talk about commutative and locally commutative squares involving maps between spaces modelled over a fixed X. In particular, how every X is modeled over yourself, we can talk about squares involving maps $f : M \to N$ and maps $g : X \to X$. This case is specially interesting. Indeed, *every map $f : M \to N$ on $\mathbf{Top}_X$ is locally conjugated to some $g : X \to X$!!!* Enough take $g = \varphi' \circ f \circ \varphi^{-1}$, where φ and φ' and charts on p and $f(p)$.

- We usually say that $\varphi' \circ f \circ \varphi^{-1}$ is the *expression* of f in the charts φ and φ'.

Globalization

- Here we will see that some concepts that make sense on some space X also make on every space modelled over X.

- Indeed, remember that every function f between spaces modelled X is locally conjugated to some function from X into yourself: the map $\varphi' \circ f \circ \varphi^{-1}$, where φ and φ' are charts. So, if $\mathcal{P}$ is local property that make sense on functions $X \to X$, then it also make sense for f. Indeed, *we can define that $\mathcal{P}$ is satisfied on f when, for every charts, $\mathcal{P}$ is satisfied on $\varphi' \circ f \circ \varphi^{-1}$.*

- **example:** differentiable functions, analytic functions, piecewise linear functions.

- We would like to define a category $\mathbf{Top}_{\mathcal{P}}$, whose objects are modelled over X and whose morphisms are maps satisfying $\mathcal{P}$. So, we need that id_M have the property $\mathcal{P}$. But id_M will satisfy $\mathcal{P}$ iff, for every charts φ and φ', the functions

$$\varphi' \circ id_M \circ \varphi^{-1} = \varphi' \circ \varphi^{-1}$$

have the property $\mathcal{P}$. We note that

$$\varphi' = (\varphi' \circ \varphi^{-1}) \circ \varphi,$$

of mode that $\varphi' \circ \varphi^{-1}$ is just the map that make the transition to the chart φ from the chart φ'. Because of this, such functions are usually called *transition functions*.

- **conclusion:** *to define* $\mathbf{Top}_{\mathcal{P}}$ *we need restrict our attention to the spaces modelled over* X *in which the transition functions have the property* $\mathcal{P}$*.*

- we observe that the isomorphisms on $\mathbf{Top}_{\mathcal{P}}$ are just homeomorphisms that have $\mathcal{P}$ and whose inverse also have $\mathcal{P}$.

- **example**: Extend the notion of differentiability on $\mathbb{R}$ to $\mathbb{R}^n$. So, talk about differentiable manifolds (**Diff**). The isomorphisms are the diffeomorphisms.

- **example:** extend the notion of class of differentiability and talk about C^k manifolds.

- **example:** extend the taylor's formula and talk about analytic manifolds.

- existence problem. Counterexamples of Michael Kervaire and Jhon Milnor at 60's.

- uniqueness problem. Counterexample of Milnor at 60's.

- **CANDY QUESTION:** *What letters of the alphabet are manifolds? What of then admits a differentiable structure? Classify all of then module isomorphisms.*

Tangent Space

- we have saw that $\mathbf{Top}_{\mathbb{R}^n}$ (and, therefore, **Diff**) does not have many limits and colimits. On the other hand, it have a very good property: *is modelled over a space that have an additional linear structure!* there is a natural functor $T : \mathbf{Diff} \to \mathbf{Vec}_{\mathbb{R}}$ that assign to every pair (M, p) a vector space TM_p and to every pair (f, p) a linear transformation $Df_p : TM_p \to TN_{f(p)}$.

- The notation is not arbitrary: we will see that TM_p and Df_p really extends the notions of tangent space and derivative.

- To associate a vector space to a manifold, we first need characterize the notion of vector using only concepts that make sense over manifolds.

- Let $D(\mathbb{R}^n)$ the set of differentiable functions $f : \mathbb{R}^n \to \mathbb{R}$. We define a *differential operator of* $D(\mathbb{R}^n)$ *on* $p \in \mathbb{R}^n$ as being a linear transformation

$$\partial_p : D(\mathbb{R}^n) \to \mathbb{R}, \quad \text{such that} \quad \partial_p(f \cdot g) = \partial_p f \cdot g(p) + f(p) \cdot \partial_p g.$$

- The collection $\mathrm{Der}_p(D(\mathbb{R}^n))$ of all these operators is a subspace of the dual $D(\mathbb{R}^n)^*$. There is a map

$$\frac{\partial}{\partial -}(p) : \mathbb{R}^n \to \mathrm{Der}(D(\mathbb{R}^n)), \quad \text{such that} \quad \frac{\partial f}{\partial v}(p) = \sum v_i \frac{\partial f}{\partial x_i}(p).$$

The fundamental fact is the following: *for every p the map $v \mapsto \frac{\partial}{\partial v}(p)$ is an isomorphism between vectors and differential operators on $\mathcal{D}(\mathbb{R}^n)$!* But we can also talk about $D(M)$ and about differential operators on this vector space. So, this characterization can be used to define vectors on manifolds.

- Indeed, fixed $p \in M$ *we define TM_p as being the vector space of all derivations of $D(M)$ on p.* Finally, given $f : M \to N$, we define

$$(Df_p)(v)(f) = v(f)(p).$$

- **Obs:** Every coordinate system induces a basis on TM_p.

Differential Forms

- Here we will use the existence of the functor $T : \mathbf{Diff} \to \mathbf{Vec}_{\mathbb{R}}$ to construct some invariants of **Diff**. These invariants are: *orientation* and *de Rham cohomology.* Both can be defined using *differential forms.*

- Remember that **Set** have products. We say that a map

$$\omega_p : TM_p \times ... \times TM_p \to \mathbb{R} \quad \text{is}$$

  - *multilinear* when it is linear on each argument.
  - *alternate* when the change of arguments imply a minus signal.

- The set of all k-multilinear forms on TM_p is a vector space $\mathrm{Mult}_k(TM_p)$. We have a natural map

$$\otimes : \mathrm{Mult}_k(TM_p) \times \mathrm{Mult}_s(TM_p) \to \mathrm{Mult}_{k+s}(TM_p),$$

called *tensor product.* It is defined by

$$(\omega_p \otimes \omega'_p)(v_1, ..., v_k, w_1, ..., w_s) = \omega_p(v_1, ..., v_k) \cdot \omega_p(v_1, ..., v_s).$$

- This map is not surjective, but their image generate $\text{Mult}_{k+s}(TM_p)$. So, there exists a basis for the space of $k+s$-multilinear forms whose elements are tensor products of k-linear forms by s-linear forms.

- Analogously, there exists a basis for the space of k-multinear forms whose elements are tensor products k different 1-multilinear forms. But 1-multinear forms is just linear maps $TM_p \to \mathbb{R}$ (that is, elements of TM_p^*). So, given a basis $\hat{e}_i$ of TM_p, we can take basis $\hat{e}^i$ of TM^* and take the set of tensor products $\hat{e}^{i_1} \otimes ... \otimes \hat{e}^{i_k}$ to get a basis for $\text{Mult}_k(TM_p)$.

- **Conclusion:** *every chart φ on p defines a basis $dx^{i_1} \otimes ... \otimes dx^{i_k}$ for* $\text{Mult}_k(TM_p)$.

- Now we observe that the set of k-multilinear alternate forms is a subspace

$$\text{Alt}_k(TM_p) \subset \text{Mult}_k(TM_p).$$

the inclusion has a left inverse

$$A : \text{Mult}_k(TM_p) \to \text{Alt}_k(TM_p).$$

- construct basis for $\text{Alt}_k(TM_p)$.

- differential forms. Vector space $\Omega^k(M)$ of differential forms.

- we have isomorphism

$$\text{Alt}_n(TM_p) \simeq \mathbb{R},$$

of mode that

$$\Omega^n(U) \simeq D(U).$$

- orientation. Volume forms. All differential form of degree n can be locally written as

$$\omega_p = f(p) \cdot dx^1 \wedge .... \wedge dx^n$$

- **Examples.** Semi-riemannian manifolds. On semi-riemannian oriented manifolds we have a natural way to construct volume forms.

Cohomology

- De Rham cohomology

Hodge Theory

- there is another way to compute the de Rham cohomology, which will explained now. The idea is the following: fixed a riemannian metric g on M, for every k it induces an inner product
$$(-,-) : \Omega^k(M) \times \Omega^k(M) \to \mathbb{R}.$$
- Relatively to this products, the exterior derivative d has an adjoint. That is, for every k there exists a linear transformation $d^\dagger : \Omega^k(M) \to \Omega^{k-1}(M)$ such that
$$(d\alpha, \beta) = (\alpha, d^\dagger \beta).$$
- So, we can define another linear transformation $\Delta : \Omega^k(M) \to \Omega^k(M)$ which measures the anticommutatibility between d and $d^\dagger$. That is,
$$\Delta = \{d, d^\dagger\} = d \circ d^\dagger + d^\dagger \circ d.$$
- The kernel of Δ is denoted by $H^k_\Delta(M)$ and their elements are the *k-harmonic forms on M*. If α in harmonic, then $d\alpha = 0$. So, the inclusion of $H^k_\Delta(M)$ into $\Omega^k(M)$ pass to quotient, inducing a natural map
$$\pi : H^k_\Delta(M) \to H^k_{dR}(M).$$
- The main theorem of Hodge theory state that this map is bijective and, therefore, an isomorphism.
- Now, to construct the inner product $(-,-)$ on each $\Omega^k(M)$, we first observe that the fixed metric g on M induces an internal product $\langle -,- \rangle_p$ on each $\mathrm{Alt}_k(TM_p)$. So, we have a map
$$\langle -,- \rangle : \Omega^k(M) \times \Omega^k(M) \to D(M), \quad \text{such that} \quad \langle \alpha, \beta \rangle(p) = \langle \alpha(p), \beta(p) \rangle_p.$$
- We define (α, β) taking into account every $\langle \alpha, \beta \rangle(p)$ in a canonical way. This can be done using the notion of *integration of n-forms*. Indeed, if ω_g is the volume form canonically induced by g, then the product $\langle \alpha, \beta \rangle \cdot \omega_g$ is a n-form, of mode that we can put
$$(\alpha, \beta) = \int_M \langle \alpha, \beta \rangle \cdot \omega_g.$$

Classification

- classification of one dimensional connected manifolds;
- classification of two dimensional compact connected oriented manifolds;
- PoincarÃ©'s conjecture;
- generalized PoincarÃ©'s conjecture.

Problems

- lie groups.
- actions, existence of pushouts by well behaved actions.

4 Dynamics

In this lecture....................

Trajectories

- A *vector field* on a manifold M is a rule X that assign a tangent vector $X(p) \in TM_p$ to each point $p \in M$. On other words, is a map $X : M \to TM$ whose image by p is exactly on the tangent space of p. This means that $\pi \circ X = id$.

- Each vector field X on M defines a *first order ordinary differential equation on* M, given by
$$x'(t) = X(x(t)).$$
The solutions for these equations are called *integral curves* of X. So an integral curve of X is just a path $x : I \to M$ such that the restriction of X to x coincides with the velocity field of x.

- We say that an integral curve x of X *comes from* $p_o \in M$ when $x(0) = p_o$. We affirm that there is an unique integral curve coming from every point of M...

  - pass the problem to $\mathbb{R}^n$ using charts;
  - $x : I \to M$ is an integral curve for X passing on p_o iff
  $$x(t) = p_o + \int_0^t X(x(s))ds.$$
  - Let $P(I; \mathbb{R}^n)$ the set of paths on some interval I of the real line. It is a certain kind of topological space called *Banach Space*. More precisely, it is a vector space is with every Cauchy sequence is convergent with respect to the norm
  $$\|x - y\| = \sup\{|x(t) - y(t)|\}.$$
  - We have an operator
  $$p_o + \int X(-)ds : P(I; \mathbb{R}^n) \to P(I; \mathbb{R}^n).$$
  - So, the integral curves of X is just the fixed points of this operator. But, on Banach spaces, every operator T satisfying
  $$\|T(x) - T(y)\| \leq \|x - y\|$$
  has an unique fixed point.
  - This is satisfied by $p_o + \int X(-)ds$ if, for example, X is C^1.

- **Example:** *vector fields on* $\mathbb{R}^n$. Remember that $TU \simeq U \times \mathbb{R}^n$, of mode that a vector field on U is a function $X : U \to \mathbb{R}^n$.

- **Example:** *a second order PVI on* $\mathbb{R}^n$ *can be view as a first order PVI with the double of coordinates.* Indeed, let $f : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^n$ be a differentiable function and

$$x''(t) = f(x(t), x'(t)) \tag{3}$$

the second order EDO associated to him. Defining $X : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ by

$$X(p, v) = (v, f(p, v)),$$

them each PVI of f can be translated on a PVI of X. The physical view of this situation is the following: in general, f is the *force* acting on a particle and $x(t)$ (resp. $x'(t)$ and $x''(t)$) is the *trajectory* (resp. velocity and acceleration) of the particle subjected by this force. So, (3) *is just the Newton's second law of motion!!* Thus, for every force (which can be explicitly dependent of the velocity) the Newton's second law also have solutions for every initial conditions.

- **Example:** *gravitational interaction.* A particular case of system included in the previous example is the following: given points $p_1, ..., p_{n-1} \in \mathbb{R}^3$ and real numbers $m, m_i, ...m_n$, we consider the function

$$f(p, v) = \sum_i G \frac{m \cdot m_i}{\|p - p_i\|^2},$$

defined on $U \times \mathbb{R}^n$, where U is the open set obtained subtracting of $\mathbb{R}^n$ the points p_i. So, the correspondent differential equation is

$$x''(t) = \sum_i G \frac{m \cdot m_j}{\|p - p_i\|^2}.$$

Physically, this system describes the motion of one particle of mass m subjected to the gravitational interaction of $n - 1$ other particles of mass $m_1, ..., m_{n-1}$, which are fixed in the points $p_1, ..., p_{n-1}$.

- **Example:** *n-body problem.* In the previous example, was supposed that the other particles stay fixed on certain points. This is not the general case..........

- two observations:

  - was seen that every 2nd order EDO on $\mathbb{R}^n$ defines, globally, a first order EDO on the double of coordinates. The reciprocal is locally true: in a sufficiently small neighborhood of the initial conditions, every first order equation on $\mathbb{R}^{2n}$ provides from the order reduction of an second order equation on $\mathbb{R}^n$. This is a direct consequence of the implicit function theorem.
  - we can talk about second order differential equations (and study theri relation with first order equations) in more general manifolds than $\mathbb{R}^n$. See Lang's book.

Integrability

- Previously we give an idea of how prove that every PVI has an unique local solution. Now, the question is: *what is the shape of this solution?* Indeed, remember that the solution was constructed by approximation methods. So, the question is *what is the limit, in terms of known function, of the sequence which produces the solution?*

- the concept of local flow. So, the question is: explicit the flow.

- a system in which we know explicitly their solutions is called *integrable*.

- **Example:** linear case. $\varphi(t,p) = e^{At}p$.

- **Example:** first order equations on 1-manifolds are locally integrable. Indeed, locally have the form $\frac{dx}{dt} = f(x)$, of mode that

$$dt = \frac{dx}{f(x)}, \quad \text{and} \quad t(x) = \int \frac{dx}{f(x)}.$$

  But, by the inverse function theorem, we always can locally invert the expression $t(x)$ to get the solution $x(t)$.

- We will not talk about *necessary* conditions for that a system be integrable. On the other hand, we give an strategy which can be used to get *sufficient* conditions for integrability.

- The idea is the following: let $X : M \to TM$ be is a vector field on a n-dimensional manifold. Now, suppose that there is a function $f : M \to \mathbb{R}$ such that

  - is a submersion. That is, $Df_p \neq 0$ for every point $p \in M$;
  - there exists a number $a \in \mathbb{R}$ such that, if x is any integral curve of X, then $f(x(t)) = a$ for every t.

- So, being f a submersion, then $f^{-1}(b)$ is a submanifold of M of dimension $n-1$. In particular, $f^{-1}(a)$ is a submanifold. But every solution of the PVI's of X are in $f^{-1}(a)$. So, we can restrict X to $f^{-1}(a)$, reducing the dimension from n to $n-1$.

- We call f an *first integral for X*. Thus, in principle, if we can give $n-1$ first integrals for a system on M, then it can be reduced to an one dimensional submanifold, being integrable. But, we observe that two *different first integrals the same kind of restriction.*

- **Conclusion:** if a vector field X on a n-manifold admits $n-1$ independent first integrals, then it is integrable by quadratures.

- physical insight to get first integrals:

- conservation laws of energy, momentum and angular momentum on each axe;
- geometric constraints;
- cinematic constraints (are essentially auxiliary EDO's).

- **Example:** *a box going down a slope.* We have two degrees of freedom. But we have many first integrals!! This means that this problem can be solved by many different perspectives!!
  - we have conservation of energy. So, using this first integral, we can integrate the system;
  - we have conservation of momentum in both axes. So, using this first integrals, we can integrate the system;
  - we have a geometric constraint: the box moving with a fix angle. So, in polar coordinates, the system is integrable.

- **Example:** *a rolling disk on a slope.*

- **Example:** *elastic collision of two balls on the plane.* If the collision is inelastic, we lost the conservation of energy and, in principle, the system can be non integrable.

- **Example:** *the general n-body problem is non integrable for $n \geq 3$.* Indeed, we have $3n$ degrees of freedom and, in principle, only $2n+1$ conservation laws. For $n = 2$, we have *six degrees of freedom*, and *five conservation laws.* So, the system is integrable by quadratures. On the other hand, for $n = 3$ we have *nine degrees of freedom* and only *seven conservation laws.*

Qualitative Theory

- How study a non-integrable system if, in principle, we cannot know explicitly their solutions? As we will see, in every case, the solutions (which really exists) give a decomposition (called *foliation*) of the manifold M by 1-dimensional manifolds oriented manifolds. This decomposition is called *phase portrait* of the system. The idea is then study properties (for example, topological properties) of the phase portrait. These properties will give *qualitative information* about the trajectories.

- The area of math which study qualitatively the PVI's is the *qualitative theory of ordinary differential equations.* So, we expect that there is a category which describes exactly this theory. The objects must be vector field on M. The mappings $f : X \to Y$ will be functions $f : M \to M$ which preserve the phase portrait. But the phase portrait is formed by oriented solutions. So, the morphisms will be functions which take solutions of X and give solutions of Y, preserving their orientations.

- Let show that, for every vector field X, the phase portrait really exists.

- we first observe that the image of every solution is an one dimensional submanifold of M. Indeed, let x be the solution passing in $p \in M$. Then $x'(t) = X(x(t))$ for all t. Suppose that there is t_o such that $x'(t_o) = 0$. So, by the translational invariance on time, $x'(t) = 0$ for every t. Thus, for every solution, or $x'(t) = 0$ for each t or $x'(t) \neq 0$ for every t. In the first case, $x(I)$ is just a point and, therefore, a zero dimensional manifold. In the other case, x is an immersion of a one dimensional manifold I on M, and then $x(I)$ is a one dimensional manifold inside M;
- by the existence of solutions, on every $p \in M$ we can build a manifold, of mode that the solutions really give a decomposition of M. By the uniqueness, the manifolds of two distinct solutions are disjoints and then the decomposition of M provided by then is also disjoint.
- finally, we can give orientations for every manifold $x(I)$ by first choosing an orientation on I and then passing it to $x(I)$ using the fact that $x'(t) \neq 0$.

- **OBS:** the solutions such that $x'(t) = 0$ for every t are called *singular solutions.* We observe that the solution of X passing on p is singular iff $X(p) = 0$. By the classification of one dimensional manifolds, the other solutions are diffeomorphic to $\mathbb{R}$ or $\mathbb{S}^1$. They are respectively called *line solutions* and *periodic solutions.*

Invariants

- We need get information about the phase portrait. Difficulty: in general, we don't know all trajectories explicitly. But we have a natural strategy! It is based on the following recipe:
  - discover *some* class of special trajectories and put them in the phase portrait (singularities and periodic orbits);
  - determines the *local behavior* of the trajectories in the neighborhood of the special trajectories;
  - now, to globalize the local information, we can determines the asymptotic behavior of the trajectories in the neighborhood of the special trajectories.
- before explicit how work on the previous strategy, we observe that (periodic, lines or singularities)
- stable and unstable manifolds
- **CANDY QUESTION:** *why stable and unstable manifolds in general are not embedding submanifolds?*

5 Morse Theory

Gradient

- Here we study the qualitative theory of a particular of differential equations: which are provided by the gradient of real functions.

- More precisely, for a given manifold M, we will show that the choose of a riemannian metric g on M establish a functor $F : \mathbf{C}_M^g \rightarrow \mathbf{DS}_M$, where $\mathbf{C}_M^g$ is the category whose morphisms are functions $f : M \rightarrow \mathbb{R}$ and whose morphisms commutative squares of diffeomorphisms which preserves the metric. That is, a morphis $f \rightarrow f'$ is just a diffeomorphism $h : M \rightarrow M$ such that $f = f' \circ h$ and

$$g_{f(p)}(Df_p(v), Df_p(w)) = g_p(v, w).$$

- Then, having constructed F, we will study the qualitative theory of the dynamical systems defined by $F(f)$.

- So, let g a riemannian metric on M . To define $F(f)$, we observe that g induces canonical isomorphisms

$$(-)^* : TM_p \rightarrow TM_p^*, \quad \text{such that} \quad v^* = g_p(v, -).$$

- Thus, we define $F(f)$ as being the vector field dual to the 1-form *df* by the previous isomorphism. It is called *gradient of f with respect to g* and usually denoted by $\mathrm{grad}_g(f)$. That is, for every p and every v,

$$Df_p(v) = \langle \mathrm{grad}_g f(p), v \rangle.$$

- Now, let $h : f \rightarrow f'$ be a morphism between two functions $f, f' : M \rightarrow \mathbb{R}$. We assert that h also defines a morphism between the gradients $\mathrm{grad}_g f$ and $\mathrm{grad}_g f'$, motivating us to put $F(h) = h$.

- We need prove that h preserves the phase portrait. That is, we need prove that, if $x : I \rightarrow \mathbb{R}$ is an integral curve of $\mathrm{grad}_g f$, then $h \circ x$ is an integral curve of $\mathrm{grad}_g f'$. Let do this.

  - First observe that x is an integral curve of $\mathrm{grad}_g f$ exactly when

$$Df_{x(t)}(v) = \langle x'(t), v \rangle \quad \text{for every} \quad t, v.$$

  - So, we need prove that

$$Df'_{(h \circ x)(t)}(w) = \langle (h \circ x)'(t), w \rangle \quad \text{for every} \quad t, w.$$

– Now, if $h : f \to f'$ is a morphism, then $f = f' \circ h$ and so $f' = f \circ h^{-1}$. Therefore,

$$\begin{aligned} Df'_{(h\circ x)(t)}(w) &= Df_{x(t)}(Dh^{-1}_{(h\circ x)(t)}(w)) \\ &= \langle x'(t), Dh^{-1}_{(h\circ x)(t)}(w)\rangle \\ &= \langle Dh_{x(t)}(x'(t)), Dh_{x(t)}(Dh^{-1}_{(h\circ x)(t)}(w))\rangle \\ &= \langle (h\circ x)'(t)), w\rangle, \end{aligned}$$

where in the third passage was used that h preserves the metric.

- **Example:** gradient forces. This NOT MEANS that the dynamics of the second law is gradient dynamics. Indeed, it is a second order equation and, when we reduces their order, the vector field lose their gradient property. With effect, it pass to be a new kind of vector fields, called *symplectic gradient*.

Phase Portrait

- Remember that to construct the phase portrait of a vector field
  - first we look for their singularities and their periodic orbits;
  - so, we search for invariants which classify the local behavior in the neighborhood of singularities and periodic orbits;
  - to globalize the information, we take the limits $t \to \pm\infty$. That is, we analyse the asymptotic behavior of the line orbits.

- We will see that *the dynamics of gradient like vector field is relatively simple*. Indeed, in this subsection, we will identify their singularities, as well show that they does not have periodic orbits. Furthermore, every line solution born and death on singularities. So, *their phase portraits are obtained simply drawing points connected by lines*. Finally, in the next subsection we show that a large class of gradient like vector fields admits a completely local classification, given by an invariant called *index*.

- First remember that, by definition, $\mathrm{grad}_g f$ is just the vector field dual to df. So, *the singularities of the gradient are exactly the critical points of f*. That is, the points p for which $Df_p = 0$.

- **Example:** singular points of conservative mechanical systems are ones in which the force is equal zero. Equivalently is the critical points of the potential. Examples: harmonic oscillator.

- Now, let show that $\text{grad}_g f$ has not periodic orbits. Let x be a non singular integral curve. So, $x'(t) \neq 0$ and, therefore,

$$Df_{x(t)}(x'(t)) = \langle x'(t), x'(t) \rangle = \|x'(t)\|^2 > 0,$$

of mode that f is strictly increasing along x, what cannot happens if x is periodic.

- Finally, let prove that the ω-limit and the α-limit of every integral curve x is a singularity.

  - if x is singular, so there are nothing to do.
  - let x non singular. So, was proven, f is strictly increasing along x. This means that
  $$f(x(t')) > f(x(t)) \quad \text{if} \quad t' > t.$$
  - now, suppose that the ω-limit of x is not a singularity. So, there is a point $p \in M$ such that $Df_p \neq 0$ and for which there is a sequence $t_k \to \infty$ such that $x(t_k) \to p$. We assert that this imply a contradiction with the increasing of f along x;
  - how $Df_p \neq 0$, it is surjective. So, $f^{-1}(p)$ is a submanifold of dimension $n-1$.
  - Being f continuous, $\lim f(x(t_k)) = f(p)$. Thus, $x(t_k)$ intersect $f^{-1}(p)$ for every large k. Therefore, for large k, we have
  $$f(x(t_{k'})) = f(x(t_k)) = f(p) \quad \text{if} \quad t_{k'} > t_k,$$
  which is the asserted contradiction.

Morse Functions

- Remember that the gradient defined a a functor $F : \mathbf{C}^g_M \to \mathbf{DS}_M$. How the critical points of f is just the singularities of their gradient, F induces a new functor

$$F : \mathbf{C}^g_{M*} \to \mathbf{DS}_{M*},$$

where $\mathbf{C}^g_{M*}$ is the category of pair (f,p), being p a critical point of f. Therefore, using F we can transfer local invariants of critical points to local invariants of singularities.

- One of these invariants is the *index of a critical point* p of f. It is defined as being the number $\text{Ind}(f,p)$ of negative eigenvalues of the hessian matrix $\frac{\partial^2 f}{\partial x_i \partial x_j}(p)$ on some coordinate system φ in p. We observe that:

  - *this number is well defined.* That is:
    * how $f \in C^1$, their hessian matrix is symmetric and, therefore, has only real eigenvalues. So, we really talk about the negative part of the eigenvalues;

    * changing the coordinates, the hessian matrix change by a similarity transformation. So, by the Silvester's law of inertia, the index is independent of the choose of coordinates.

  - *it really is an invariant.* Indeed, if $h : f \to f'$ is an equivalence, then................

- There is special interest when the critical point p *is not degenerated.* This means

$$\det \frac{\partial^2 f}{\partial x_i \partial x_j}(p) \neq 0$$

or, equivalently, that the eigenvalues of the hessian in every coordinate system is every positive or negative. Indeed, in these cases *the index is sufficient to classify the function in the neighborhood of* p.

- The idea is the following: observe that, for every coordinate system φ in p, we have the Taylor formula

$$\begin{aligned}(f \circ \varphi)(q) &= (f \circ \varphi)(p) + \sum_i \frac{\partial f}{\partial x_i}(p) x_i + \sum_{i,j} \frac{\partial^2 f}{\partial x_i \partial x_j}(p) x_i x_j + \mathcal{O}(x^3) \\ &= (f \circ \varphi)(p) + \sum_{i,j} \frac{\partial^2 f}{\partial x_i \partial x_j}(p) x_i x_j + \mathcal{O}(x^3).\end{aligned}$$

- In particular, we can choose a coordinate in which the hessian matrix is diagonal with unitary eigenvalues, and then, if λ is the index of p,

$$(f \circ \varphi)(q) = (f \circ \varphi)(p) - x_1^2 - ... - x_\lambda^2 + x_{\lambda+1}^2 + ... + x_n^2 + \mathcal{O}(x^3).$$

- So, if we look in a sufficiently small neighborhood, the additional terms $\mathcal{O}(x^3)$ are irrelevant and then f is conjugated by the quadratic form we can write

$$(f \circ \varphi)(p) - x_1^2 - ... - x_\lambda^2 + x_{\lambda+1}^2 + ... + x_n^2,$$

which is totally determined by the index.

- At this moment, was not explicitly used the hypothesis that p is non degenerated. But they appear implicitly in the last passage, when we intuitively remove the additional terms $\mathcal{O}(x^3)$. Indeed, the formal argument is the following:

  - observe that the hessian matrix of f is just the jacobian matrix of the gradient $\mathrm{grad}_g f$. So, p is non degenerated exactly when $D(\mathrm{grad}_g f)_p$ is an isomorphism. So, by the non degenerated hypothesis, we can use the inverse function theorem, which asserts the existence of the chart φ in which the higher order terms $\mathcal{O}(x^3)$ can be despised.

- Morse functions.
- A critical point with maximal index is called an *atractor*. Saddle point.
- **Example:** height function on the torus. We have four critical points of index 0, 1, 1 and 2.
- Density and stability of Morse functions.

Decomposition

- Here we will given an idea of a fundamental result: *every manifold is a CW-complex.*
- First we observe that, by certain results called *stable and unstable manifolds theorems*, for every singular point p of index λ, the sets $W^s(p)$ and $W^u(p)$ are respectively homeomorphic to the interiors of the disks $\mathbb{D}^\lambda$ and $\mathbb{D}^{n-\lambda}$.
- So, every Morse function $f : M \to \mathbb{R}$ decomposes as union of interior of disks: one for each critical point. But these decompositions are not necessarily CW-complex. The classical counterexample is just the height function on the torus:
- **Example:** fsfsfs
- On the other hand, we observe that a little of the...
- every manifold has a decomposition on CW-complex. So, for them we have uniqueness of well behaved homology and cohomology theories
- the necessity of transversally between the stable and unstable manifolds. Morse-Smale functions.

Morse Homology

- counting curves as a way to construct invariants. The same idea can be applied to construct a topological string theory, as we will see on the final lecture.
- let f be a Morse-Smale function and let p and q critical points of index λ_p and λ_q. Then $W^u(p)$ and $W^u(q)$ intersect traversally. So,

$$W^u(p) \cap W^s(q) = \{x(t), \lim_{t\to-\infty} x(t) = p \quad \text{and} \lim_{t\to\infty} x(t) = q\}$$

$$C_k(f) = \text{free abelian group generated by the set of critical points of index } k.$$

$$\partial_k : C_k(f) \to C_{k-1}(f)$$

$$\partial_k(\sum a_i p_i) = \sum a_i \partial_k(p_i) = \sum a_i \sum n(p_i, q)q.$$

Existence of Equilibrium

- so, there exists to many n^2

6 Quantum Mechanics

Abstract Description

- In the modern abstract viewpoint there are two kind of physical theories: *classical theories* and *quantum theories*. Here, we will describe briefly classical and quantum theories for particles.

- both kinds of theories are based on the notion of "configuration". Indeed, a *configuration* is just a map $\varphi : \Sigma \to M$ of a 1-manifold Σ to a m-manifold M. The idea is think Σ as a *abstract motion* of an arbitrary 0-dimension object along the time (which is one dimensional). The map $\varphi : \Sigma \to M$ is a certain king of embedding on the manifold M, and so we can say that *a configuration is just a motion on a background ambient.*

- **Obs:** By the classification theorem of one dimensional manifolds, an arbitrary particle can describe exactly in one of the following kinds of motions: $\Sigma = I$ or $\Sigma = \mathbb{S}^1$. The first case represent a particle that parts of a determined point and return to a different point. The second, on other hand, corresponds to a particle which returns to the initial point. Here we will look specifically for the case in which $\Sigma = I$.

- Roughly speaking, a *classical theory* is just one which *select the possible configurations.* One the other hand, a *quantum theory* is one in which *all configurations is possible, but some of them are "more possible" than others.* In other words, classical theories are mechanisms (or schemes) which choose the "true trajectory" and quantum theories are mechanisms which give probabilities to all of them. Thus, classical theories are *deterministic* and quantum theories are *probabilistic.*

- In classical or quantum theories, we also need talk about the *time evolution* of the system. Indeed, this information is about how the

Concrete Description

- In general, a classical theory select the possible configurations by the exigence that they must satisfy some differential equation, called *equation of motion.* Furthermore, is just this equation that will describes the time evolution of the system. For example, this is the case of the second Newton's law on classical mechanics.

- On other hand, in quantum theories we associate to every $t \in I$ an unitary operator $Z_t : \mathcal{H} \to \mathcal{H}$ on a complex vector space $\mathcal{H}$, called *time evolution operator*, of mode that we have *translational invariance* $Z_{t+t'} = Z_t \circ Z_{t'}$. So, by a theorem of Stone, there exists another hermitean operator H, called *Hamiltonian operator of the system*, such that $Z_t = \exp(-iHt)$.

- In general we proceed taking $\mathcal{H}$ as the space $L^2(M;\mathbb{C},\mu)$ of square integrable functions $\psi : M \to \mathbb{C}$, with the canonical internal product

$$\langle \psi, \xi \rangle = \int \psi(p)\overline{\xi(p)} d\mu.$$

To define the operator Z_t we first associate a certain kind of "weight" $w(\varphi)$ to every configuration φ, and then, fixing $p_o, p \in M$, we suppose that we have a notion of "*integration over all configurations φ satisfying $\varphi(0) = p_o$ and $\varphi(t) = p$*", like

$$K(p, p_o) = \int w(\varphi) D\varphi. \tag{4}$$

So, we put

$$Z_t(\psi)(x) = \int K(p, p_o)\psi(p_o) d\mu$$

The problem with is this process is that, in general, *the integral* (4) *is not well defined on the mathematical viewpoint*. Special cases will be discussed on future lectures.

- **Example:** *quantization*. In this case, $w(\varphi) = \exp(iS[\varphi])$.

Functoriality

- Here we observe that a quantum theory of a particle is a certain kind of functor Z. Indeed, remember that we need construct a complex vector space $\mathcal{H}$ and, for every $t \in I$, an unitary operator Z_t satisfying $Z_{t+t'} = Z_t \circ Z_{t'}$. So our functor must take values in the category $\mathbf{Vec}_{\mathbb{C}}$.

- What is their domain? We observe that, by the translational invariance, define Z_t for every t is equivalent to define $Z_{t_2;t_1}$ for every interval $[t_2, t_1]$ inside I: enough take $Z_{t_2;t_1} = Z_{t_2 - t_1}$. But such interval is just a 1-manifold whose boundary is the coproduct of the zero dimensional manifolds t_1 and t_2. Because of this, we say that $[t_1, t_2]$ is a *cobordism* between t_1 and t_2.

- So, Z must take cobordisms inside I and return operators on $\mathbf{Vec}_{\mathbb{C}}$. This means that cobordisms will be the *morphisms* in the domain $\mathbf{Cob}_1$ of Z. The objects of $\mathbf{Cob}_1$ can be taken simply as the zero manifolds $t \in I$. The composition will be given by the gluing along common boundary: if $[t_1, t_2]$ and $[t_2, t_3]$ are cobordisms inside I, then $[t_3, t_2] \circ [t_2, t_1] = [t_3, t_1]$. To see that these definitions really makes $Z : \mathbf{Cob}_1 \to \mathbf{Vec}_{\mathbb{C}}$ a functor, observe that

$$\begin{aligned} Z([t_3, t_2] \circ [t_2, t_1]) &= Z([t_3, t_1]). \\ &= Z_{t_3;t_1} \\ &= Z_{t_3 - t_1} \end{aligned}$$

$$
\begin{aligned}
&= Z_{t_3-t_2+(t_2-t_1)} \\
&= Z_{t_3-t_2} \circ Z_{t_2-t_1} \\
&= Z([t_3,t_2]) \circ Z([t_2,t_1])
\end{aligned}
$$

and, being $[t,t]$ the identity on $\mathbf{Cob}_1$ for every t, that

$$Z([t,t]) = Z_{t;t} = Z_{t-t} = Z_0 = id_{\mathcal{H}}.$$

Susy

- We have commented that exists the notion of *quantization* of a classical system. This means that every classical concept and every classical property has a quantum analogous. The reciprocal, however, is not valid. Indeed, quantum particles can have some properties which, in principle, does not admits any classical analogous.

- A roughly justification for this fact is the following: if on one hand classical systems are realized on the category of manifolds, which is badly behaved, on another hand quantum systems are realized on $\mathbf{Vec}_{\mathbb{C}}$, which is very nice behaved category (for example, it is complete, cocomplete and abelian). So, we can require certain properties for quantum systems which, in principle, does not make sense for classical system.

- **Example:** if $Z : \mathbf{Cob}_1 \to \mathbf{Vec}_{\mathbb{C}}$ defines a quantum system, we can require that the correspondent space $\mathcal{H}$ admits a coproduct decomposition and we can talk about operators on $\mathcal{H}$ which preserves (or mix) some of the parts. For example, we can lead with *supersymmetric quantum mechanics*. These are quantum systems for which exists operators Q and $Q^\dagger$ such that $\{Q, Q^\dagger\} = 2H$.

- **Example:** as we will see later, *spinning quantum particles have supersymmetry.*

- vacuum states.

Hodge Theory

- Here, we will see that for every differentiable manifold M with a fixed riemannian metric g, there is a realization of susy quantum mechanics in which vacuum states computes harmonic forms (and, therefore, de Rham cohomology) of M.

- Indeed, the idea is observe that the supercharges Q and $Q^\dagger$ must satisfy $Q^2 = 0$ and $Q^\dagger = 0$ on $\mathcal{H}$, like the exterior derivative d and their adjoint $d^\dagger$ on $\Omega^*(M)$. So, the idea is take
$$\mathcal{H} = \Omega^*(M), \quad Q = d, \quad Q^\dagger = d^\dagger,$$
of mode that
$$H = \frac{1}{2}\{Q, Q^\dagger\} = \frac{1}{2}\{d \circ d^\dagger + d^\dagger \circ d\} = \frac{1}{2}\Delta,$$

implying that the vacuum states of the quantum system are really realized as the harmonic forms on M.

- Happens that $\mathcal{H}$ must be complex and $\Omega^*(M)$ is initially real. But this is not a problem: for every ring R and every subring S, we have a natural functor

$$- \otimes_S R : \mathbf{Mod}_S \to \mathbf{Mod}_R.$$

  So, $\Omega^*(M) \otimes_{\mathbb{R}} \mathbb{C}$ is a complex vector space. It also has a inner product, given by a notion of integration of differentiable forms. So, we can consider it as $\mathcal{H}$ and the correspondent operators $d \otimes_{\mathbb{R}} \mathbb{C}$ and $d^\dagger \otimes_{\mathbb{R}} \mathbb{C}$, which satisfy exactly the same properties that d and $d^\dagger$, as Q and $Q^\dagger$.

Morse Theory

- the addition of a potential which is a Morse function don't change the cohomology. Use this to show that Morse homology is isomorphic to De Rham cohomology and, therefore, to recover the Morse inequalities.

Floer Homology

- here we will see that

7 Cobordism

8 K-Theory

9 Geometry

- newtonian approach to motion.
- liouville integrability as a integrability condition for 1-dimensional distribuition. Generalization for EDP's and frobenius theorem.
- spinning quantum particles have supersymmetry on the worldline.

10 Spin

11 Gauge Theory

12 Geometric Quantization

13 QFT

- conformal invariance
- quantum anomalies
- string theory and the justification for the critical dimensions.

14 TQFT

15 Functorial Quantization